Explicit Nash Equilibrium in the Simplified Linear-Utility Two-Player Model

1 Overview and Purpose

This document derives closed-form expressions for the Nash equilibrium mixing parameters λ_L^* and λ_H^* in a highly simplified two-player model that captures the core macro dynamics of the original N-player stochastic job-swap system.

We deliberately relax the Friedman–Savage S-shaped utility to linear utility $u(y) = y$ (pure risk neutrality), while preserving the essential behavioral asymmetry through class-specific quadratic risk premiums: - Low-rank player receives a positive premium for lottery allocation (mimicking risk-seeking / convexity effect). - High-rank player receives a negative premium (mimicking risk-aversion / concavity effect).

The result is a tractable model with explicit solutions that reproduce persistent inequality, higher lottery exposure at the bottom, and stability at the top — without stochastic draws or nonlinear utility curvature.

2 Model Setup

Two players exist: low-rank (L) and high-rank (H).

Each receives current income m_w at the beginning of period t: $m_L = R\mu, \quad m_H = R(1-\mu)$, $\mu \in (0, 0.5), \quad$ R ¿ $0(totalresourcescale)$.

Each player chooses a mixing parameter $\lambda_w \in [0, 1]$: - fraction λ_w of current income allocated to a shared lottery pool, - fraction $1 - \lambda_w$ allocated to upskilling (safe improvement channel).

Net income y_w (which is consumed entirely this period and becomes next-period income $m_w^{t+1} = y_w^t$, no saving) is:

$$
\begin{aligned}
y_w = m_w + \eta(1-\lambda_w)m_w - C_w(\lambda_w) - \lambda_w m_w \\
+ s_w \kappa \Pi + \rho_w(\lambda_w)
\end{aligned}
$$

where the components are defined as follows:

- $\Pi = \lambda_L m_L + \lambda_H m_H \quad$ (total size of shared lottery pool)
- $s_w = \dfrac{\lambda_w m_w}{\Pi} \quad$ (player's proportional share of the pool)
- $\kappa \approx 1 \quad$ (mean return per unit contributed to the pool; fairness parameter)
- $\eta \in (0, 1) \quad$ (efficiency of upskilling investment)

- $C_w(\lambda_w) \approx b\eta(1-\lambda_w)m_w$ (linearized upskilling cost, $b > 1$ approximates original power-law friction)
- $\rho_w(\lambda_w) = p_w\lambda_w^2$ (quadratic risk-premium approximation)
  - $p_L > 0$ $\rightarrow$ extra deterministic bonus for higher λ_L (proxies convexity / risk-seeking)
  - $p_H < 0$ $\rightarrow$ deterministic penalty for higher λ_H (proxies concavity / risk-aversion)

Because utility is linear, each player simply maximizes y_w with respect to their own λ_w (treating the other's choice as fixed).

3 Derivation of Closed-Form Equilibrium

3.1 Low-rank player's first-order condition

Differentiate y_L with respect to λ_L (ignoring small higher-order terms in share derivative for tractability):

$$\frac{\partial y_L}{\partial \lambda_L} \approx -\eta m_L + b\eta m_L - m_L + \kappa m_L \cdot (\text{share adjustment term}) + 2p_L\lambda_L$$

After simplification (dominant linear terms + quadratic premium, dropping complex share cross-derivative):

$$\frac{\partial y_L}{\partial \lambda_L} \approx (-\eta + b\eta - \kappa + 1)m_L + 2p_L\lambda_L$$

Set the derivative to zero:

$$(-\eta + b\eta - \kappa + 1)m_L + 2p_L\lambda_L = 0$$

Solve for λ_L^*:

$$\lambda_L^* = \frac{m_L(\eta - b\eta + \kappa - 1)}{-2p_L} = \frac{m_L(b\eta - \eta - \kappa + 1)}{2p_L}$$

(since $p_L > 0$ and the numerator is typically positive when upskilling costs dominate).

3.2 High-rank player's first-order condition

The derivation is symmetric:

$$\lambda_H^* = \frac{m_H(b\eta - \eta - \kappa + 1)}{2p_H}$$

Because $p_H < 0$, the denominator is negative, which tends to produce small (or negative) values for λ_H^*, often clipped at the boundary.

3.3 Bounding the solutions

Final equilibrium values are bounded to the feasible interval:

$$\lambda_w^* = \max\big(0, \min\big(1, \frac{m_w(b\eta - \eta - \kappa + 1)}{2p_w}\big)\big) \qquad \text{for } w = L, H$$

4 Interpretation of the Closed-Form Expressions

- The common factor $(b\eta - \eta - \kappa + 1)$ captures the net marginal benefit of moving one unit of income from upskilling to the lottery pool.
  - Higher $b\eta$ (more expensive/costly upskilling) $\rightarrow$ favors lottery.
  - Higher κ (more generous lottery return) $\rightarrow$ favors lottery.
- Division by $2p_w$ scales the response to the risk-premium strength:
  - $p_L > 0 \rightarrow$ larger premium reduces λ_L^* (quadratic saturation effect).
  - $p_H < 0 \rightarrow$ larger $|p_H|$ increases λ_H^* toward 1 (but usually still clips at 0).
- The income term m_w in the numerator means richer players receive proportionally larger absolute responses — but the sign of p_w dominates direction.

5 Numerical Illustration

Baseline parameters: $R = 1$, $\mu = 0.2 \implies m_L = 0.2$, $m_H = 0.8$, $\eta = 0.9$, $b = 2$, $\kappa = 1$, $p_L = 0.5$, $p_H = -0.3$.

Common term: $b\eta - \eta - \kappa + 1 = 1.8 - 0.9 - 1 + 1 = 0.9$

Equilibrium:

$$\lambda_L^* = \frac{0.2 \times 0.9}{2 \times 0.5} = \frac{0.18}{1} = 0.180$$
$$\lambda_H^* = \frac{0.8 \times 0.9}{2 \times (-0.3)} = \frac{0.72}{-0.6} = -1.20 \quad \rightarrow \boxed{0}$$

The high player fully avoids the lottery ($\lambda_H^* = 0$), consistent with "sticky top" behavior in the original model.

6 Remarks

These closed-form expressions are simple, interpretable, and computationally trivial — ideal for analytical comparative statics, spreadsheet experiments, or teaching.

The corner solution for λ_H^* is typical when the negative premium is strong; it faithfully reproduces the original model's qualitative pattern of low-rank churning and high-rank stability.

To obtain interior solutions for both players, reduce $|p_H|$ or introduce a small positive bias in the high player's lottery return.